

Review

Emerging novel therapies for steroid-refractory acute graft-versus-host disease: recent advances and future directions

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Acute graft-versus-host disease (aGVHD) is a significant complication of allogeneic hematopoietic cell transplantation (allo-HCT), characterized by immune-mediated tissue damage from donor immune cells. Standard treatment of aGVHD involves systemic corticosteroids, but many patients do not respond adequately, with 30%–50% of patients being steroid refractory (SR-aGVHD) leading to poor outcomes. This highlights the need for effective second-line therapies. Ruxolitinib, a JAK1/2 inhibitor, has emerged as a key treatment, demonstrating superior overall response rates in SR-aGVHD compared to best available therapies. However, resistance and intolerance to ruxolitinib necessitate exploring novel and combination therapies such as apraglutide, neihulizumab, and other targeted agents, which have shown promising results in clinical trials. Additionally, nonpharmacologic approaches, including microbiotherapy, mesenchymal stromal cells, or alpha-1 antitrypsin, are also being investigated for their immunomodulatory potential. The future direction is to develop personalized treatment strategies that incorporate biomarkers and diverse therapeutic modalities. The aim is to enhance disease management, aiming for sustainable control and improved quality of life in patients facing the challenges of SR-aGVHD.

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Introduction

Graft-versus-host disease (GVHD) is a major complication following allogeneic hematopoietic cell transplantation (allo-HCT) [1]. It arises when donor immune cells recognize the host's tissues as foreign, leading to immune-mediated tissue damage. GVHD is classified into acute (aGVHD) and chronic forms based on the features of the disease. Therefore, while aGVHD typically presents within the first 100 days post-transplant, it can also occur beyond day 100 and can involve the skin, liver, and/or gastrointestinal (GI) tract.

The mainstay of treatment for aGVHD is systemic corticosteroids, which exert their effects by broadly suppressing the immune response. However, a significant proportion of patients, ranging from 30% to 50%, fail to respond adequately to steroid therapy, a condition termed steroid-refractory aGVHD (SR-aGVHD). In addition, some patients, despite responding to steroids, may become dependent on them; a condition known as steroid-dependent (SD) aGVHD. This high rate of steroid resistance or dependence poses a significant clinical challenge, as SR- and SD-aGVHD are associated with poor prognosis, substantial morbidity, and increased mortality [2,3].

Steroid-refractory acute graft-versus-host disease

SR-aGVHD is defined by the progression of GVHD after 3 days or absence of clinical improvement after 7 days of adequate doses of systemic corticosteroids (typically prednisone at ≥ 2 mg/kg/day or equivalent) [4]. Other criteria include worsening symptoms after an initial response or the need for escalating steroid doses. SD-aGVHD is usually defined as the inability to taper corticosteroid therapy below 0.5 mg/kg/day. SR- or SD-aGVHD necessitates the introduction of second-line therapies.

Ruxolitinib

Ruxolitinib, a selective JAK1/2 inhibitor, has emerged as a pivotal therapy for SR-aGVHD due to its ability to inhibit the cytokine-mediated inflammatory pathways central to GVHD pathophysiology. By blocking the JAK-STAT signaling pathway, ruxolitinib reduces the pro-

duction of pro-inflammatory cytokines, such as interleukin (IL)-6, IL-12, and interferon- γ [5]. The efficacy of ruxolitinib in SR-aGVHD was demonstrated in the landmark REACH2 trial, a randomized phase III study comparing ruxolitinib to the best available therapy (BAT). Ruxolitinib achieved significantly higher overall response rates (ORRs) and durable responses at day 28 compared to BAT [6]. Based on these results, ruxolitinib was approved as the first- and second-line treatments for SR-aGVHD.

While ruxolitinib represents a significant advance, a subset of patients remains resistant or intolerant to the drug, highlighting an unmet need for alternative or adjunctive therapies. In the REACH1 [7] and REACH2 [6] trials, 45% and 38% of patients did not have a complete response (CR) or partial response (PR) at day 28, respectively. Moreover, in the REACH2 trial, the ORR at day 56 after initiation of therapy decreased to 40%. These results clearly suggest the unmet need for patients with GVHD who did not respond at day 28 or who worsened afterward. In order to identify patients that fail ruxolitinib before day 28, we proposed in 2020, a working definition of ruxolitinib-resistance aGVHD [4], which was defined as (1) progression of GVHD compared with baseline after at least 5–10 days of treatment with ruxolitinib based either on objective increase in stage/grade or new organ involvement; (2) lack of improvement in GVHD (PR or better) compared with baseline after 14 days of treatment with ruxolitinib; or (3) loss of response, defined as objective worsening of GVHD determined by increase in stage, grade, or new organ involvement at any time after initial improvement [4]. The prognosis of ruxolitinib-refractory aGVHD is very dismal with 31% of ruxolitinib-resistant patients dying before initiation of additional therapy and for patients receiving such therapy, an ORR to next therapy was 36%, leading to a median survival of 21 days [8].

Therefore, there is an urgent need for innovative drugs to address the issue of steroid- and ruxolitinib-resistant aGVHD, drugs which should possess both efficacy and durability. However, when ruxolitinib is administered early (considered as initial treatment failure), it becomes challenging to assess any benefits from novel investigational methods. In the case of SR-aGVHD, discussion is ongoing as to whether future studies should compare treatments with ruxolitinib alone or explore the potential of combination therapies [9].

Emerging agents for steroid-refractory acute graft-versus-host disease

Novel therapies are being developed to address the limitations of current treatments and provide additional options for refractory cases. The relevant clinical trials are summarized in Table 1 and mechanisms of action illustrated in Figure 1.

Monoclonal antibodies

Apraglutide

Glucagon-like peptide 2 (GLP-2) is an enteroendocrine hormone produced by intestinal L cells, which has been proven in a preclinical study to protect and regenerate Paneth cells and intestinal stem cells and thus has the potential to improve clinical outcomes in GI aGVHD [10]. The STARGAZE phase 2 trial (NCT05415410) tested the use of the GLP-2 analog apraglutide in combination with ruxolitinib for SR lower-GI aGVHD. The recently released results showed that apraglutide was well-tolerated, with day 28 all organ ORR and CR rates of 58.1% and 25.8%, respectively, and lower-GI ORR and CR rates of 54.8% and 29%, respectively. Notably, durable overall response from day 56 to day 91 was 41.9% for the all-organ ORR and 45.2% for the lower-GI ORR [11].

Neihulizumab

Neihulizumab (ALTB-168) is a novel immune checkpoint agonistic antibody that binds to human CD162 (PSGL-1) and leads to downregulation of activated T cells. In a phase I trial studying single-agent neihulizumab in SR-aGVHD, neihulizumab was well tolerated and achieved a day 28 ORR of 69% in cohort 1 (single-dose 3 mg or 6 mg) and 50% in cohort 2 (first dose at 6 mg/kg followed by weekly doses of 4 mg/kg for 3 weeks). Of note, in the latter cohort, 13 SR-aGVHD patients who received additional treatment ($n = 11$, ruxolitinib, $n = 2$ unknown) after steroid and before neihulizumab therapy achieved a higher ORR than those who did not (67% vs 33%), supporting further investigation of neihulizumab as an agent in ruxolitinib-refractory aGVHD [12].

Begelomab

Begelomab, a monoclonal antibody directed against CD26, an antigen expressed on activated T cells, was tested in 69 patients (61 of whom had grade III-IV aGVHD) within two prospective studies ($n = 28$) and one compassionate use study ($n = 41$) for patients with SR aGVHD who did not receive ruxolitinib. Of note, patients treated in the compassionate use study had received 1–4 previous lines of treatment for aGVHD. Begelomab achieved a day 28 ORR of 75% in the two prospective studies and 61% in the compassionate use patients, with low CR rates being 11% and 12%, respectively, and a 1-year overall survival (OS) of 50 for the prospective studies and 33% for the compassionate use patients [13]. A prospective, phase II/III, randomized clinical study to compare begelomab versus conventional treatment in SR-aGVHD was terminated early due to lack of accrual (NCT02411084).

Tocilizumab

Tocilizumab, an IL-6 receptor blocker, which is FDA approved to treat severe active rheumatoid arthritis and

Table 1**Clinical trials on novel therapies for steroid-refractory aGVHD.**

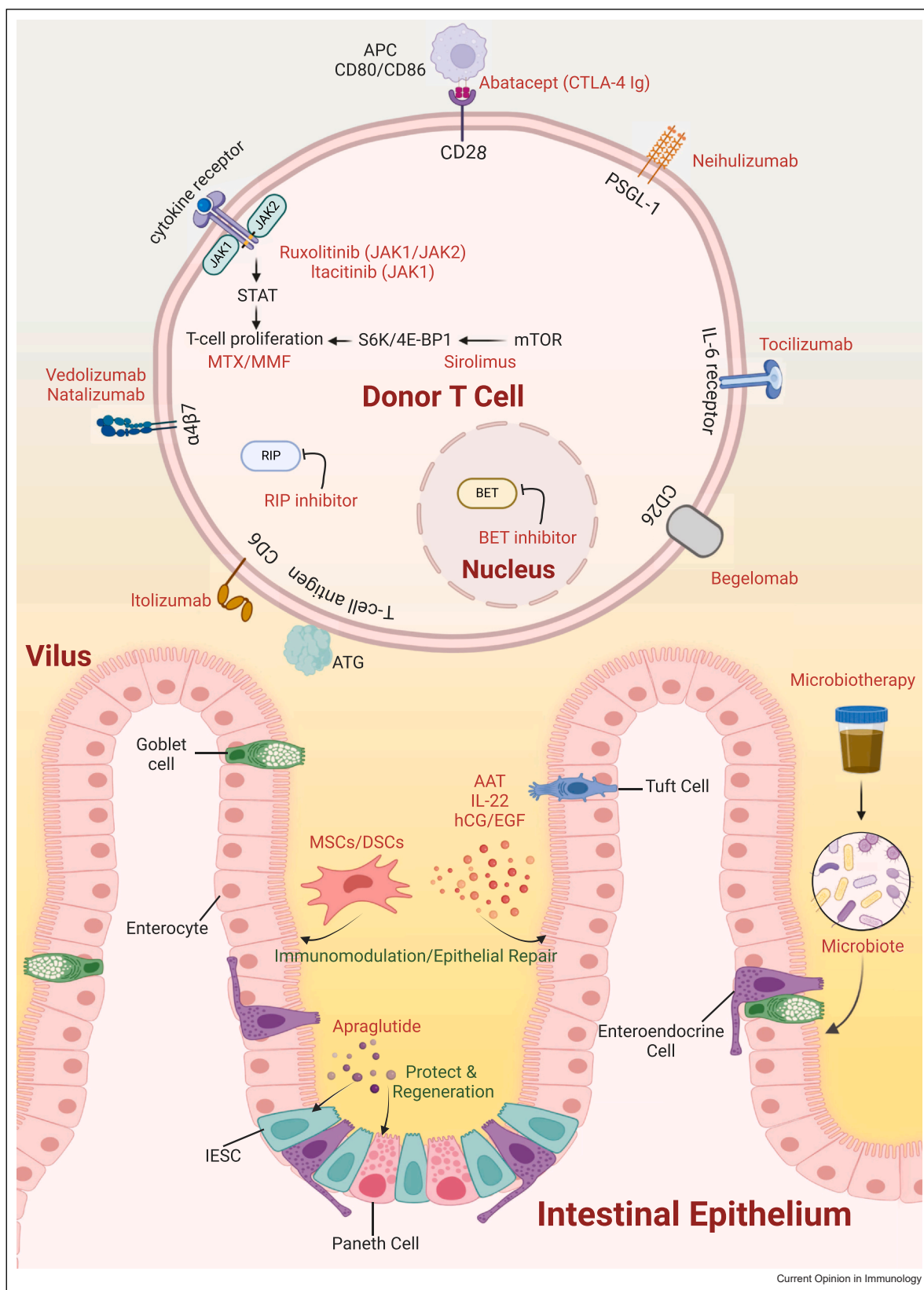
Drug/treatment	Clinical trial name and phase	Clinical trial number	Results
Trials on SR-aGVHD			
Ruxolitinib [6]	REACH2, Phase 3	NCT02913261	Day 28 ORR: 62%; CR: 34%
Ruxolitinib [7]	REACH1, Phase 2	NCT02953678	Day 28 ORR: 54.9%; CR: 26.8%
Apraglutide [11]	STARGAZE, Phase 2	NCT05415410	Day 28 ORR: 58.1%, CR: 25.8%; Day 28 lower-GI ORR: 54.8%, CR: 29%; Day 28 ORR: 69%
Neihulizumab [12]	Phase 1	NCT03327857	Day 28 ORR:
Begelomab [13]	Phase 1/2	EudraCT No. 2007-005809-21 AND EudraCT No. 2012-001353-19	75% in the prospective cohort 61% in compassionate use patients Day 28 CR: 11% in the prospective cohort 12% in compassionate use patients
FMT (MaaT013) [32,33]	Phase 2	NCT03359980	HERACLES study Day 28 GI ORR rate: 38%; Compassionate use patients Day 28 ORR rate: 49%; Steroid and ruxolitinib refractory Day 28 GI ORR: 59%; 1-year OS: 72% Day 28 ORR: 83%
Mesenchymal stromal cells (MSCs) ^a [39]	Phase 3	NCT02241018	Median failure-free survival: 11.3 months
hCG/EGF combination [46]	Phase 2	NCT02525029	Day 28 ORR: 73%; CR: 50%;
hCG/EGF combination [45]	Phase 1	NCT02525029	Day 28 CR: 54%
Alpha-1 antitrypsin (AAT) [49]	Phase 2	NCT01700036	Day 28 ORR: 65%; CR: 35% Responses sustained in 73% of patients without intervening immunosuppression.
Vedolizumab [18]	Phase 2a	NCT02993783	Failed to meet primary efficacy end point
Microbiotherapy (MaaT013)	Phase 3	NCT04769895	Results awaited
Decidual stromal cells (DSCs)	Phase 1/2	NCT02172937	Results awaited
Vedolizumab [18]	Phase 2a	NCT02993783	Terminated due to failure to meet the primary efficacy end point
Tocilizumab	Phase 1/2	NCT01475162	Terminated for safety by the data and safety monitoring board
Tocilizumab and itacitinib combined	Phase 1	NCT04070781	Terminated for lack of recruitment
Trials on aGVHD prophylaxis			
Vedolizumab [19]	Phase 3	NCT03657160	180-day lower-GI aGVHD-free survival: 85.5%
FMT [34,35]	Phase 2	NCT03678493	FMT was administered upon neutrophil recovery 4-month infection density: FMT, 0.74 vs placebo, 0.91
FMT (MaaT013)	Phase 2b	NCT05762211	Results awaited
AAT	Phase 2/3	NCT03805789	Results awaited
Trials on aGVHD first-line treatment			
Natalizumab [20]	Single-Arm Phase 2	NCT02176031	Day 28 ORR for new onset acute GI aGVHD: 57%
Natalizumab [21]	Phase 2	NCT02133924	Day 28 ORR for new onset grade II-III aGVHD: 60% No significant difference with corticosteroids alone
Itolizumab [17]	EQUATOR Phase 3	NCT05263999	Trial recruiting
Itolizumab [16]	EQUATE Phase 1b/2	NCT03763318	Day 29 CR: 55%, ORR: 68% for newly diagnosed grade III-IV acute GI GVHD
IL-22 (F-652) [36]	Phase 2	NCT02406651	Day 28 ORR for new onset lower-GI aGVHD: 70%
AAT [50]	Phase 3	NCT04167514	Day 28 ORR AAT, 60% vs placebo, 45%
hCG/EGF ^b	Phase 1/2	NCT05123040	Results awaited

^a In combination with basiliximab and calcineurin inhibitor.^b in combination with ruxolitinib and de-escalated corticosteroids.

has had remission-inducing efficacy in systemic juvenile idiopathic arthritis, was also tested in SR-aGVHD. A small ($n = 6$) preliminary study reported a day 56 ORR of 66% in SR-aGVHD [14]. Another retrospective analysis

included 16 patients with SR lower-GI aGVHD and reported a 62.5% CR rate after tocilizumab use (8 mg/kg every 2 weeks until achievement of CR or progression) [15]. A prospective study (NCT01475162) that evaluated

Figure 1



(caption on next page)

Novel therapeutics to steroid-refractory acute GVHD. APC, antigen-presenting cell; ATG, anti-thymocyte globulin; CTLA-4, cytotoxic T-lymphocyte antigen-4; JAK, Janus kinase; MMF, mycophenolate mofetil; mTOR, mammalian target of rapamycin; MTX, methotrexate; PSGL-1, P-selectin glycoprotein ligand 1; STAT, signal transducer and activator of transcription protein.

tocilizumab in SR-aGVHD was terminated early for safety reasons by the data and safety monitoring board, while another study (NCT04070781) that evaluated tocilizumab combined with itacitinib for SR-aGVHD was terminated early due to under-recruitment.

Itolizumab

Itolizumab is a monoclonal antibody against the co-stimulatory receptor CD6 that blocks its interaction with activated leukocyte cell adhesion molecule, thereby inhibiting effector T-cell (Teff) activity and trafficking to target organs. A phase 1b/2 study evaluated the addition of itolizumab to corticosteroids for first-line treatment of patients with grade III-IV aGVHD. This study showed that at day 29 of treatment, 55% experienced a CR with an ORR of 68% [16]. A placebo-controlled, phase 3 clinical trial (NCT05263999) is therefore ongoing to assess the efficacy of itolizumab combined with corticosteroids as a first-line therapy for aGVHD [17]. Itolizumab is not currently being investigated in SR-aGVHD.

Vedolizumab

A monoclonal antibody targeting integrin $\alpha 4\beta 7$, specifically affecting gut-homing lymphocytes, was developed for the treatment of moderate–severe ulcerative colitis and Crohn's disease. In allo-HCT patients, it was first investigated in a phase 2a clinical trial that randomized two doses of vedolizumab (300 vs 600 mg) for the treatment of SR GI aGVHD [18]. This study failed to meet the primary efficacy end point and was terminated early [18]. An international phase 3 randomized, placebo-controlled trial was then conducted and showed that the addition of vedolizumab to a calcineurin inhibitor combined with methotrexate or mycophenolate mofetil for GVHD prophylaxis resulted in a higher day 180 lower-GI aGVHD-free survival than placebo in unrelated donor allo-HCT [19]. Overall, vedolizumab may have a greater advantage for GVHD prophylaxis than treatment via prevention of T lymphocyte migration towards the GI tract.

Natalizumab

Natalizumab is a humanized monoclonal antibody that binds the $\alpha 4$ subunit of $\alpha 4\beta 7$ integrins and blocks their adhesion to their counter receptors on endothelial cells. Natalizumab has only been investigated in first-line treatment of aGVHD. In a first single-arm phase II study of natalizumab with corticosteroids for first-line treatment of acute GI GVHD ($n = 21$), the day 28 ORR rate was 57%, indicating potential efficacy [20]. Nevertheless, in a larger phase II study, 75 patients were treated and compared with 150 well-matched controls

from the MAGIC database, and no significant differences in overall or CR were observed between patients treated with natalizumab plus corticosteroids and those treated with corticosteroids alone for new onset severe aGVHD [21]. Therefore, given the lack of available data on natalizumab in SR-aGVHD and negative results in the first-line setting, this study is no longer investigating aGVHD, even though prophylactic use may be interesting given that natalizumab and vedolizumab share a target.

Targeted treatment

Receptor-interacting protein kinase 1 inhibitor

Receptor-interacting protein kinase 1 (RIP1) plays a key role in inflammation and cell death in response to tissue damage and pathogen recognition. RIP1 inhibition was shown to improve immune reconstitution and reduce GVHD mortality while preserving graft-versus-leukemia effects in a preclinical study in mouse models of aGVHD [22]. A phase I clinical trial (NCT05673876) was conducted to study GDC-8264, an RIP1 inhibitor, in combination with standard of care for the treatment of aGVHD but was terminated early in 2024 by the sponsor for business reasons, and no other RIP1 inhibitors are currently being developed for aGVHD.

Bromodomain and extra-terminal domain inhibitors

Bromodomain and extra-terminal domain (BET) proteins influence inflammatory gene transcription and therefore represent a potential target to mitigate inflammation central to aGVHD pathogenesis. Inhibition of BET proteins disrupts the expression of key inflammatory genes, inactivated macrophages, and T-cells and exhibits significant anti-inflammatory properties [23]. BET inhibitors have shown efficacy in GVHD suppression in several preclinical studies in murine models of aGVHD [23–25]. Encouraged by these results, a phase Ib/II trial clinical trial was initiated to test the BET inhibitor PLX51107 (NCT04910152) in treating SR-aGVHD but was terminated early due to sponsor decision in Jan 2025.

Immune modulation and nonpharmacologic interventions

Beyond pharmacologic therapies, immune modulation strategies and supportive interventions are gaining attention for their potential to improve outcomes in SR-aGVHD.

Microbiotherapy

It is well established that loss of bacterial diversity is associated with GVHD-related outcomes after allo-HCT [26]. Domination of *Enterococcus* after allo-HCT is

associated with an increased GVHD-related mortality [27]. In contrast, butyrate-producing bacteria, such as *Blautia*, have anti-inflammatory homeostatic roles and are associated with a decreased GVHD-related mortality [28]. Fecal microbiota transfer (FMT) was investigated to treat GI aGVHD, with a first report in 2016 of four patients with SR GI aGVHD using a healthy relative FMT donor; three patients achieved a CR and one a PR [29]. Based on these results, several retrospective studies using third-party healthy transplant donors confirm [30,31] that FMT may be effective and safe in treating SR-aGVHD. Nevertheless, FMT essentially relies on stool bank availability, and development of standardized microbiome-based therapeutics is essential to improve accessibility. We therefore evaluated in a phase 2 study the use of the first-in-human pooled allogeneic fecal microbiota, MaaT013, an off-the-shelf readily available product to treat grade III-IV SR GI aGVHD. The day 28 GI ORR rate was 38% in 24 prospectively enrolled patients [32]. In addition, 154 patients were treated as part of a compassionate use program in Europe, the day 28 ORR was 49% [33]. Furthermore, 58 patients received MaaT013 as third-line treatment after steroid and ruxolitinib failure, in those patients, the day 28 GI ORR was 59% and the all organ ORR, translated to a 1-year OS of 72% in surviving patients. A prospective phase III study was then conducted to evaluate MaaT013 in SR- and ruxolitinib-refractory aGVHD (NCT04769895). FMT was also evaluated in combination with ruxolitinib in SR GI aGVHD in a retrospective analysis conducted by Liu et al. The day 28 ORR rate was 71.4% [31]. Given its satisfactory safety profile, FMT at the time of neutrophil recovery to prevent aGVHD was also examined in a phase 2 trial (NCT03678493), which was safe and ameliorated intestinal dysbiosis [34,35]. A large phase IIb randomized study (PHOEBUS) is evaluating the efficacy of MaaT033 in improving survival of adult allo-HCT patients by restoring microbiome diversity (NCT05762211).

Interleukin-22

IL-22 is a tissue-protective IL-10-family cytokine, which promotes epithelial regeneration and induces innate antimicrobial molecules. F-652, a novel recombinant human (rh) IL-22 molecule with extended half-life was tested in a phase 2 study with systemic corticosteroids as initial treatment for acute lower-GI GVHD ($n = 27$). A day 28 ORR rate of 70% was achieved, and responders had improvement of GVHD-associated dysbiosis [36]. Since IL-22 does not directly regulate enteric bacteria as do antibiotics or even prebiotics, it may represent a new class of ‘peribiotics’ that enhance mucosal health and indirectly influence the microbiome. Instead of merely increasing immunosuppression, IL-22 therapy offers a potential strategy to integrate immunosuppression with targeted tissue repair, promoting recovery from immune-

mediated damage. Therefore, IL-22 warrants further investigation in aGVHD particularly in SR patients, in combination with other approaches including FMT.

Mesenchymal stromal cells

Mesenchymal stromal cells (MSCs) are multipotent progenitor cells with multipotency and immunomodulatory properties. They have been used successfully in the treatment of tissue repair and autoimmune diseases, including aGVHD. With several retrospective studies showing efficacy of MSCs in SR-aGVHD [37,38], a prospective trial comparing second-line therapy based on basiliximab and a calcineurin inhibitor, with or without MSCs for SR-aGVHD showed a 28-day ORR of 83% and a failure-free survival of 11.3 months, which is superior to control [39]. A study recently reported the use of a random-donor allogeneic MSC preparation, MSC-FFM, available via Hospital Exemption in Germany for treatment of SR- and ruxolitinib refractory aGVHD. MSC-FFM was administered at the recommended dose of 1–2 million MSCs/kg body weight in four once-weekly doses to 156 patients, including 33 children. In adults, the ORRs at day 28 and day 60 were 45.5% and 48.4%, respectively. The OS rates at 6, 12, and 24 months were 47%, 35%, and 30%, respectively [40]. With their multifaceted functions and well-established safety profile, MSCs are being extensively investigated in aGVHD [41].

Decidual stromal cells

Decidual stromal cells (DSCs) are stromal cells derived from the placenta and may inhibit alloreactive T-cells better than other sources of stromal cells from *in vitro* studies. In a retrospective study, 38 patients with grade I-IV aGVHD (including 25 patients with SR-aGVHD) were treated with DSCs; an ORR rate of 81% and a CR rate of 42% was achieved [42]. Prospective randomized trials comparing DSCs with best available immunosuppressive therapy are needed to prove efficacy [43].

Human chorionic gonadotropin/epidermal growth factor combination

Urinary-derived human chorionic gonadotropin (uhCG) with supplemental epidermal growth factor (EGF) may facilitate resolution of aGVHD through immunomodulation and aid in epithelial repair [44]. In a phase 1 study, 26 patients were treated with uhCG/EGF in addition to standard immunosuppression ($n = 13$ in first line and $n = 13$ in second line), leading to a day 28 CR rate of 62% in first line and 54% in the second line [45]. In another phase 2 trial where hCG/EGF was also added to second-line immunosuppression for SR-aGVHD ($n = 22$), 50% of the patients achieved CR with an ORR of 73% [46]. uhCG/EGF is being investigated in a phase I/II study to establish the lowest possible recommended phase 2 dose of corticosteroids in

conjunction with ruxolitinib and uhCG/EGF for first-line treatment of high-risk aGVHD (NCT05123040).

Alpha-1 antitrypsin

α 1-Antitrypsin (AAT) is a 52-kDa circulating protease inhibitor produced by the liver that inactivates several serine proteases from neutrophils and macrophages and protects tissues from proteolytic degradation. The effects of AAT on GVHD treatment are associated with reduction in inflammatory cytokines and alterations in the ratios of Teff to regulatory T cells (Tregs) and levels of damage-associated molecular patterns [47,48]. In a phase 2 clinical trial examining AAT administration for 40 patients with SR-aGVHD, 65% and 35% of the patients achieved an ORR and CR, respectively, by day 28, and the responses were sustained at day 60 in 73% of patients without intervening immunosuppression [49]. Based on these promising results, a phase 3 randomized study is being conducted to compare the CR and PR on day 28 postrandomization between AAT + steroids versus placebo + steroids for first-line treatment of aGVHD with high-risk features (NCT04167514) [50]. In the primary ITT analysis, the day 28 ORR was 60% for patients assigned to AAT compared to 45% for placebo ($p = 0.034$), which confirms the promising results of the phase 3 studies. Nevertheless, longer term follow-up is needed for further assessment of NRM, OS, and other critical end points. In addition, AAT is also being investigated in an ongoing phase 2/3 trial for aGVHD prevention in patients undergoing an allo-HCT (NCT03805789).

Conclusion

Significant progress has been made in the management of SR-aGVHD, with ruxolitinib serving as a cornerstone therapy. However, challenges remain for patients resistant to ruxolitinib, emphasizing the need for novel therapies. Advances in monoclonal antibodies, emerging agents targeting specific pathways, and non-pharmacologic approaches offer hope for improving outcomes in this challenging population.

Future research should focus on personalized treatment strategies, integrating biomarkers to guide therapy selection and combining pharmacologic and non-pharmacologic approaches to achieve durable disease control and improved quality of life for patients with SR-aGVHD.

Declaration of Competing Interest

MM reports grants and lecture honoraria from Janssen, Sanofi, Maat Pharma, and JAZZ Pharmaceuticals, lecture honoraria from Celgene, Amgen, BMS, Takeda, and Pfizer, and grants from Roche, all outside the submitted work. FM reports lecture honoraria from Therakos/Mallinckrodt, BMS, MSD, Sanofi,

Novartis, AstraZeneca, and JAZZ Pharmaceuticals, all outside the submitted work.

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
- of outstanding interest

1. Malard F, Holler E, Sandmaier BM, Huang H, Mohty M: **Acute graft-versus-host disease**. *Nat Rev Dis Prim* 2023, **9**:27.
2. Holtan SG, Yu J, Paranagama D, Tang J, Choe HK, Naim A, Joachim Deeg H, Galvin J: **Disease progression, hospital readmissions, and clinical outcomes for patients with steroid-refractory acute graft-versus-host disease: a multicenter, retrospective study**. *Bone Marrow Transpl* 2022, **57**:1399-1404.
3. Martin PJ: **How I treat steroid-refractory acute graft-versus-host disease**. *Blood* 2020, **135**:1630-1638.
4. Mohty M, Holler E, Jagasia M, Jenq R, Malard F, Martin P, Socie G, Zeiser R: **Refractory acute graft-versus-host disease: a new working definition beyond corticosteroid refractoriness**. *Blood* 2020, **136**:1903-1906.
5. Maffini E, Giaccone L, Festuccia M, Brunello L, Buondonno I, Ferrero D, Boccadoro M, Dellacasa C, Busca A, Novero D, et al.: **Ruxolitinib in steroid refractory graft-vs.-host disease: a case report**. *J Hematol Oncol* 2016, **9**:67.
6. Zeiser R, von Bubnoff N, Butler J, Mohty M, Niederwieser D, Or R, Szer J, Wagner EM, Zuckerman T, Mahuzier B, et al.: **Ruxolitinib for glucocorticoid-refractory acute graft-versus-host disease**. *N Engl J Med* 2020, **382**:1800-1810.
7. Jagasia M, Perales MA, Schroeder MA, Ali H, Shah NN, Chen YB, Fazal S, Dawkins FW, Arbushites MC, Tian C, et al.: **Ruxolitinib for the treatment of steroid-refractory acute GVHD (REACH1): a multicenter, open-label phase 2 trial**. *Blood* 2020, **135**:1739-1749.
8. Abedin S, Rashid N, Schroeder M, Romee R, Nauffal M, Alhaj Moustafa M, Kharfan-Dabaja MA, Palmer J, Hogan W, Hefazi M, et al.: **Ruxolitinib resistance or intolerance in steroid-refractory acute graft-versus-host disease - a real-world outcomes analysis**. *Br J Haematol* 2021, **195**:429-432.
9. Zhao Y, Wu H, Shi J, Luo Y, Li X, Lan J, Ni W, Lu Y, Chen L, Tan Y, et al.: **Ruxolitinib combined with etanercept induce a rapid response to corticosteroid-refractory severe acute graft vs host disease after allogeneic stem cell transplantation: results of a multi-center prospective study**. *Am J Hematol* 2020, **95**:1075-1084.
10. Norona J, Apostolova P, Schmidt D, Ihlemann R, Reischmann N, Taylor G, Kohler N, de Heer J, Heeg S, Andrieux G, et al.: **Glucagon-like peptide 2 for intestinal stem cell and Paneth cell repair during graft-versus-host disease in mice and humans**. *Blood* 2020, **136**:1442-1455.
11. Zeiser R, Choe H, Adelman H, Youssef NN, Ereemeeva A, Collin F, Chen Y-B: **Safety and efficacy of the glucagon-like peptide 2 (GLP-2) analog apraglutide in patients with steroid-refractory gastrointestinal acute graft-versus-host disease (aGVHD) in combination with best available therapy: results from a multicenter, randomized, single-blind, proof-of-concept, phase 2 Stargaze trial**. *Blood* 2024, **144**:100.
12. Abedin S, Hamadani M, Holtan SG, Schiller GJ, Gallogly M, Waller EK, Kosuri S, Abhyankar SH, Anand SM, Gooptu M, et al.: The results of STARGAZE, the first prospective trial of a GLP-2 analog in SR lower-GI aGVHD, show that GLP-2 analog apraglutide was well tolerated and efficacious when added to ruxolitinib in the second-line setting.

- Neihulizumab (ALTB-168) in patients with steroid-refractory acute graft-versus-host-disease (SR-aGVHD) or treatment-refractory acute graft-versus-host-disease (TR-aGVHD).** *Blood* 2022, **140**:10495-10497.
13. Bacigalupo A, Angelucci E, Raiola AM, Varaldo R, Di Grazia C, Gualandi F, Benedetti E, Risitano A, Musso M, Zallio F, *et al.*: **Treatment of steroid resistant acute graft versus host disease with an anti-CD26 monoclonal antibody-Begelomab.** *Bone Marrow Transpl* 2020, **55**:1580-1587.
 14. Drobyski WR, Pasquini M, Kovatovic K, Palmer J, Douglas Rizzo J, Saad A, Saber W, Hari P: **Tocilizumab for the treatment of steroid refractory graft-versus-host disease.** *Biol Blood Marrow Transpl* 2011, **17**:1862-1868.
 15. Ganetsky A, Frey NV, Hexner EO, Loren AW, Gill SI, Luger SM, Mangan JK, Martin ME, Babushok DV, Drobyski WR, *et al.*: **Tocilizumab for the treatment of severe steroid-refractory acute graft-versus-host disease of the lower gastrointestinal tract.** *Bone Marrow Transpl* 2019, **54**:212-217.
 16. Koreth J, Ritz J, Chinn L, Ng C, Acevedo LM, Chu D, Fung M, Rothman J, Connelly S, Thomas D, *et al.*: **Itolizumab, a novel targeted anti-cd6 therapy, in combination with corticosteroids, is well-tolerated, with rapid pharmacodynamic and clinical response in newly diagnosed acute graft-versus-host disease.** *Blood* 2021, **138**.
 17. Koreth J, Kean L, Chinn L, Acevedo LM, Bandman O, Rothman J, Connelly S, Thomas D, Cutler C: **Pb2196: equator: a pivotal phase 3 double-blind, randomized, placebo-controlled study of itolizumab in combination with corticosteroids for the initial treatment of acute graft-versus-host disease.** *HemaSphere* 2022, **6**:2066-2067.
 18. Floisand Y, Schroeder MA, Chevallier P, Selleslag D, Devine S, Renteria AS, Mohty M, Yakoub-Agha I, Chen C, Parfionovas A, *et al.*: **A phase 2a randomized clinical trial of intravenous vedolizumab for the treatment of steroid-refractory intestinal acute graft-versus-host disease.** *Bone Marrow Transpl* 2021, **56**:2477-2488.
 19. Chen YB, Mohty M, Zeiser R, Teshima T, Jany O, Maertens J, Purtill D, Chen J, Cao H, Rossiter G, *et al.*: **Vedolizumab for the prevention of intestinal acute GVHD after allogeneic hematopoietic stem cell transplantation: a randomized phase 3 trial.** *Nat Med* 2024, **30**:2277-2287.
 20. Kekre N, Kim HT, Hofer J, Ho VT, Koreth J, Armand P, Nikiforow S, Gooptu M, Romee R, Alyea EP, *et al.*: **Phase II trial of natalizumab with corticosteroids as initial treatment of gastrointestinal acute graft-versus-host disease.** *Bone Marrow Transpl* 2021, **56**:1006-1012.
 21. Al Malki MM, London K, Baez J, Akahoshi Y, Hogan WJ, Etra A, Choe H, Hexner E, Langston A, Abhyankar S, *et al.*: **Phase 2 study of natalizumab plus standard corticosteroid treatment for high-risk acute graft-versus-host disease.** *Blood Adv* 2023, **7**:5189-5198.
 22. Prado-Acosta M, Jeong S, Utrero-Rico A, Goncharov T, Webster JD, Holler E, Morales G, Dellepiane S, Levine JE, Rothenberg ME, *et al.*: **Inhibition of RIP1 improves immune reconstitution and reduces GVHD mortality while preserving graft-versus-leukemia effects.** *Sci Transl Med* 2023, **15**:eadf8366.
 23. Sun Y, Wang Y, Toubai T, Oravec-Wilson K, Liu C, Mathewson N, Wu J, Rossi C, Cummings E, Wu D, *et al.*: **BET bromodomain inhibition suppresses graft-versus-host disease after allogeneic bone marrow transplantation in mice.** *Blood* 2015, **125**:2724-2728.
 24. Copsel SN, Lightbourn CO, Barreras H, Lohse I, Wolf D, Bader CS, Manov J, Kale BJ, Shah D, Brothers SP, *et al.*: **BET bromodomain inhibitors which permit Treg function enable a combinatorial strategy to suppress GVHD in pre-clinical allogeneic HSCT.** *Front Immunol* 2018, **9**:3104.
 25. Snyder KJ, Choe HK, Gao Y, Sell NE, Braunreiter KM, Zitzer NC, Neidemire-Colley L, Kalyan S, Dorrance AM, Keller A, *et al.*: **Inhibition of bromodomain and extra terminal (BET) domain activity modulates the IL-23R/IL-17 axis and suppresses acute graft-versus-host disease.** *Front Oncol* 2021, **11**:760789.
 26. Shono Y, van den Brink MRM: **Gut microbiota injury in allogeneic haematopoietic stem cell transplantation.** *Nat Rev Cancer* 2018, **18**:283-295.
 27. Stein-Thoeringer CK, Nichols KB, Lazrak A, Docampo MD, Slingerland AE, Slingerland JB, Clurman AG, Armijo G, Gomes ALC, Shono Y, *et al.*: **Lactose drives Enterococcus expansion to promote graft-versus-host disease.** *Science* 2019, **366**:1143-1149.
 28. Shono Y, Docampo MD, Peled JU, Perobelli SM, Velardi E, Tsai JJ, Slingerland AE, Smith OM, Young LF, Gupta J, *et al.*: **Increased GVHD-related mortality with broad-spectrum antibiotic use after allogeneic hematopoietic stem cell transplantation in human patients and mice.** *Sci Transl Med* 2016, **8**:339ra71.
 29. Kakhana K, Fujioka Y, Suda W, Najima Y, Kuwata G, Sasajima S, Mimura I, Morita H, Sugiyama D, Nishikawa H, *et al.*: **Fecal microbiota transplantation for patients with steroid-resistant acute graft-versus-host disease of the gut.** *Blood* 2016, **128**:2083-2088.
 30. van Lier YF, Davids M, Haverkate NJE, de Groot PF, Donker ML, Meijer E, Heubel-Moene F, Nur E, Zeerleder SS, Nieuwdorp M, *et al.*: **Donor fecal microbiota transplantation ameliorates intestinal graft-versus-host disease in allogeneic hematopoietic cell transplant recipients.** *Sci Transl Med* 2020, **12**:eaa8926.
 31. Liu Y, Zhao Y, Qi J, Ma X, Qi X, Wu D, Xu Y: **Fecal microbiota transplantation combined with ruxolitinib as a salvage treatment for intestinal steroid-refractory acute GVHD.** *Exp Hematol Oncol* 2022, **11**:96.
 32. Malard F, Loschi M, Huynh A, Cluzeau T, Guenounou S, Legrand F, Magro L, Orvain C, Charbonnier A, Panz-Klapuch M, *et al.*: **Pooled allogeneic faecal microbiota MaaT013 for steroid-resistant gastrointestinal acute graft-versus-host disease: a single-arm, multicentre phase 2 trial.** *EClinicalMedicine* 2023, **62**:102111.
- In this phase 2 trial, we applied MaaT013, an off-the-shelf readily available product to treat grade III-IV SR GI aGVHD. The day 28 GI ORR rate was 38% in 24 prospectively enrolled patients. Compared with single donors, MaaT013 is characterized by higher microbial richness and reduced variability across batches.
33. Malard F, Loschi M, Cluzeau T, Devillier R, Legrand F, Charbonnier A, Lebon D, Huynh A, Borel C, Méar J-B, *et al.*: **Pooled fecal allogenic microbiotherapy for refractory gastrointestinal acute graft-versus-host disease: results from early access program in Europe.** *Blood* 2024, **144**.
- MaaT013 was also tested in the early access program in Europe, it was effective for refractory GI-aGVHD in patients having previously received ruxolitinib. Interestingly, the response to MaaT013 correlates with increased OS, suggesting a favorable benefit-risk profile.
34. Rashidi A, Ebadi M, Rehman TU, Elhusseini H, Kazadi D, Halaweish H, Khan MH, Hoeschen A, Cao Q, Luo X, *et al.*: **Randomized double-blind phase II trial of fecal microbiota transplantation versus placebo in allogeneic hematopoietic cell transplantation and AML.** *J Clin Oncol* 2023, **41**:5306-5319.
 35. Rashidi A, Ebadi M, Rehman TU, Elhusseini H, Kazadi D, Halaweish H, Khan MH, Hoeschen A, Cao Q, Luo X, *et al.*: **Potential of fecal microbiota transplantation to prevent acute GVHD: analysis from a phase II trial.** *Clin Cancer Res* 2023, **29**:4920-4929.
 36. Ponce DM, Alousi AM, Nakamura R, Slingerland J, Calafiore M, Sandhu KS, Barker JN, Devlin S, Shia J, Giralt S, *et al.*: **A phase 2 study of interleukin-22 and systemic corticosteroids as initial treatment for acute GVHD of the lower GI tract.** *Blood* 2023, **141**:1389-1401.
 37. Dotoli GM, De Santis GC, Orellana MD, de Lima Prata K, Caruso SR, Fernandes TR, Rensi Colturato VA, Kondo AT, Hamerschlag N, Simoes BP, *et al.*: **Mesenchymal stromal cell infusion to treat steroid-refractory acute GVHD III/IV after hematopoietic stem cell transplantation.** *Bone Marrow Transpl* 2017, **52**:859-862.
 38. Ball LM, Bernardo ME, Roelofs H, van Tol MJ, Contoli B, Zwaginga JJ, Avanzini MA, Conforti A, Bertina A, Giorgiani G, *et al.*: **Multiple infusions of mesenchymal stromal cells induce sustained remission in children with steroid-refractory, grade III-IV acute graft-versus-host disease.** *Br J Haematol* 2013, **163**:501-509.

39. Zhao K, Lin R, Fan Z, Chen X, Wang Y, Huang F, Xu N, Zhang X, Zhang X, Xuan L, *et al.*: **Mesenchymal stromal cells plus basiliximab, calcineurin inhibitor as treatment of steroid-resistant acute graft-versus-host disease: a multicenter, randomized, phase 3, open-label trial.** *J Hematol Oncol* 2022, **15**:22.

In this randomized phase 3 trial, the authors observed that MSCs plus second-line treatments achieved higher OR rate and longer median failure-free survival compared with second-line treatments for SR aGVHD. cGVHD incidence was also decreased.

40. Bonig H, Verbeek M, Herhaus P, Braitsch K, Beutel G, Schmid C, Muller N, Bug G, Doring M, von Stackelberg A, *et al.*: **Real-world data suggest effectiveness of the allogeneic mesenchymal stromal cells preparation MSC-FFM in ruxolitinib-refractory acute graft-versus-host disease.** *J Transl Med* 2023, **21**:837.
41. Zhao Y, Luo Y, Shi J, Yu J, Liu L, Lai X, Fu H, Ye Y, Li N, Jiang E, *et al.*: **Sequential umbilical cord derived mesenchymal stromal cells (MSCs) for the third-line salvage treatment of steroid-refractory acute GvHD: a multicenter, open label, phase 1b/2 trial.** *Blood* 2023, **142**.
42. Ringden O, Baygan A, Remberger M, Gustafsson B, Winiarski J, Khoein B, Moll G, Klingspor L, Westgren M, Sadeghi B: **Placenta-derived decidual stromal cells for treatment of severe acute graft-versus-host disease.** *Stem Cells Transl Med* 2018, **7**:325-331.
43. Ringden O, Sadeghi B: **Placenta-derived decidual stromal cells: a new frontier in the therapy of acute graft-versus-host disease.** *Stem Cells* 2024, **42**:291-300.
44. Schumacher A, Heinze K, Witte J, Poloski E, Linzke N, Woidacki K, Zenclussen AC: **Human chorionic gonadotropin as a central regulator of pregnancy immune tolerance.** *J Immunol* 2013, **190**:2650-2658.
45. Holtan SG, Hoeschen AL, Cao Q, Arora M, Bachanova V, Brunstein CG, Miller JS, Rashidi A, Slungaard A, Ustun C, *et al.*: **Facilitating resolution of life-threatening acute GVHD with human chorionic gonadotropin and epidermal growth factor.** *Blood Adv* 2020, **4**:1284-1295.
46. Holtan SG, Hoeschen A, Cao Q, Ustun C, Betts BC, Jurdi NE, Maakaron J, Rashidi A, Miller JS, Wagner JE, *et al.*: **Phase II, open-label clinical trial of urinary-derived human chorionic gonadotropin/epidermal growth factor for life-threatening acute graft-versus-host disease.** *Transpl Cell Ther* 2023, **29**:509 e501-509 e508.
47. Tawara I, Sun Y, Lewis EC, Toubai T, Evers R, Nieves E, Azam T, Dinarello CA, Reddy P: **Alpha-1-antitrypsin monotherapy reduces graft-versus-host disease after experimental allogeneic bone marrow transplantation.** *Proc Natl Acad Sci USA* 2012, **109**:564-569.
48. Marcondes AM, Karopongse E, Lesnikova M, Margineantu D, Welte T, Dinarello CA, Hockenbery D, Janciauskiene S, Deeg HJ: **Alpha-1-Antitrypsin (AAT)-modified donor cells suppress GVHD but enhance the GVL effect: a role for mitochondrial bioenergetics.** *Blood* 2014, **124**:2881-2891.
49. Magenau JM, Goldstein SC, Peltier D, Soiffer RJ, Braun T, Pawarode A, Riwe MM, Kennel M, Antin JH, Cutler CS, *et al.*: **alpha(1)-Antitrypsin infusion for treatment of steroid-resistant acute graft-versus-host disease.** *Blood* 2018, **131**:1372-1379.
50. Magenau J, Hamadani M, Bai L, Martens M, Abedin S, Arai S, Ponce D, El Jurdi N, Choe H, Oshima M, *et al.*: **A randomized, double-blind, placebo-controlled phase III trial of alpha-1-antitrypsin for treatment of high risk acute Gvhd (BMT CTN 1705).** *Transplant Cell Ther Meet ASTCT CIBMTR* 2025,.